

P.E.S.COLLEGE OF ENGINEERING, MANDYA-571401

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)



MINI PROJECT REPORT ON

“VIRTUAL PLC USING ARDUINO”

Submitted in partial fulfilment of the requirement for the award of degree

Bachelor of Engineering In

ELECTRICAL AND ELECTRONICS

ENGINEERING

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DEPARTMENT OF ELECTRICAL & ELECTRONICS

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CERTIFICATE

This is to certify that, **Nihal N (4PS23EE023), Rohith H B (4PS23EE034), Rohith R Gowda (4PS23EE035), Sai Prathiksha (4PS23EE037)** have successfully completed Mini Project work entitled **“VIRTUAL PLC USING ARDUINO”** in partial fulfillment towards **Bachelor of Engineering in Electrical and Electronics Engineering of P.E.S. college of Engineering, Mandya, VTU Belagavi** during the year **2025-2026**. It is certified that the report meets all the guidelines laid out by the department.

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ABSTRACT

Industrial automation systems widely use Programmable Logic Controllers (PLCs) for controlling machines and monitoring industrial processes. However, conventional PLCs are often expensive and require specialized hardware and software, making them less suitable for educational institutions, research laboratories, and small-scale automation projects. To overcome these limitations, this project proposes a Virtual PLC using Arduino, which serves as a low-cost and flexible alternative to traditional PLC systems.

The proposed system utilizes an Arduino Uno microcontroller as the central control unit to perform PLC-like operations such as input scanning, logic execution, timing control, and output switching. Push-button inputs are used for Start, Stop, Mode Selection, and Reset functions, while a four-channel relay module controls external electrical loads. The system supports both Manual Mode, where loads are controlled directly, and Automatic Mode, where loads operate sequentially with predefined time delays to simulate industrial automation processes.

Status indication LEDs provide real-time information about power availability, operating mode, run condition, and fault status. The compact design, ease of programming using Arduino IDE, and low implementation cost make the system suitable for educational, research, and small-scale industrial applications. The project demonstrates that an Arduino-based Virtual PLC can effectively replicate the basic functions of a conventional PLC while offering greater affordability, flexibility, and scalability for future automation developments.

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INTRODUCTION

In recent years, industrial automation has become increasingly important for improving productivity, efficiency, and reliability in manufacturing and control systems. Programmable Logic Controllers (PLCs) are widely used in industries for controlling machines and automation processes due to their robustness and real-time operation. However, conventional PLCs are often expensive and may not be affordable for educational institutions, research laboratories, and small-scale automation projects. To address this limitation, microcontroller-based systems such as Arduino can be used to develop a Virtual PLC that performs basic PLC functions at a significantly lower cost.

This project focuses on the design and implementation of a Virtual PLC using the Arduino microcontroller. The Arduino acts as the central processing unit and performs PLC-like operations such as input monitoring, logic execution, timing control, and output switching. The system utilizes push-button switches for Start, Stop, Mode Selection, and Reset operations, providing a simple and user-friendly interface. Relay modules are connected to the Arduino to control external electrical loads safely and effectively.

The Virtual PLC supports both Manual and Automatic modes of operation. In Manual mode, the connected loads can be controlled directly by the user, while in Automatic mode, the loads operate sequentially according to predefined logic and timing conditions, simulating industrial automation processes. Status indication LEDs are incorporated to display power availability, operating mode, run status, and fault conditions, allowing real-time monitoring of the system.

A regulated power supply ensures stable operation of the Arduino and associated hardware components. The compact design, ease of programming through the Arduino IDE, and low implementation cost make the system highly suitable for educational purposes, prototype development, and small-scale industrial automation applications. Overall, the project demonstrates that an Arduino-based Virtual PLC can serve as a practical, flexible, and economical alternative to traditional PLC systems while providing a platform for learning and experimentation in industrial automation.

Literature Survey

1. Programmable Logic Controllers (PLCs)

Programmable Logic Controllers (PLCs) have been extensively used in industrial automation because of their reliability, flexibility, and ability to perform real-time control operations. According to W. Bolton, PLCs operate through a continuous scan cycle consisting of input monitoring, program execution, diagnostics, and output updating. PLCs are widely implemented in manufacturing industries for machine control, process automation, and monitoring applications. However, the high cost of PLC hardware and software limits their use in educational and small-scale automation projects.

2. PLC Programming and Industrial Automation

Frank D. Petruzella explained that PLCs provide efficient control solutions through ladder logic programming and modular input/output architecture. Traditional PLC systems are highly reliable but require specialized hardware and proprietary programming tools. Researchers have therefore explored low-cost alternatives capable of performing PLC-like operations while maintaining ease of programming and flexibility.

3. Arduino-Based Embedded Control Systems

Arduino microcontrollers have emerged as popular platforms for automation and control applications due to their low cost, open-source architecture, and ease of programming. According to Banzi and Shiloh, Arduino provides a simple environment for developing embedded control systems capable of handling digital inputs, outputs, timing operations, and communication functions. These features make Arduino a suitable platform for implementing Virtual PLC systems.

4. Microcontroller-Based Industrial Automation

Several studies have demonstrated the use of microcontrollers in industrial automation applications such as motor control, relay switching, process monitoring, and sequential operation control. Researchers have shown that microcontroller-based systems can effectively replace conventional relay logic circuits while providing greater flexibility through software-based modifications and upgrades.

5. Virtual PLC Systems

Virtual PLC technology aims to replicate the core functions of traditional PLCs using software and embedded hardware platforms. Various researchers have developed Virtual PLC models capable of executing logic control, timing functions, and process sequencing.

6. Industry 4.0 and Smart Automation

The development of Industry 4.0 has accelerated the adoption of intelligent and interconnected automation systems. Studies by Jay Lee and other researchers emphasize the integration of computation, communication, and physical processes to achieve smart manufacturing. Arduino-based Virtual PLCs support these concepts by enabling programmable control, sensor integration, and communication capabilities for modern automation applications.

7. Relay-Based Automation and Sequential Control

Many industrial processes require the sequential operation of motors, pumps, conveyors, and lighting systems. Researchers have developed Arduino-based relay control systems capable of implementing time-based sequencing and process control. These systems demonstrate that Arduino can effectively perform automation tasks while providing safe isolation between low-voltage control circuits and high-power loads through relay modules.

8. Human-Machine Interface and Monitoring Systems

Recent research has focused on improving user interaction in automation systems through push buttons, LCD displays, and status indication LEDs. Such interfaces provide real-time monitoring of system conditions and simplify operation. Arduino-based automation systems commonly incorporate these features to enhance usability and system reliability.

9. Research Gap and Motivation

Although traditional PLCs offer excellent reliability and industrial performance, their high cost and complexity restrict their usage in educational and small-scale automation environments. Existing Arduino-based automation projects often focus only on simple relay switching and basic control operations. There is a need for a low-cost Virtual PLC system capable of performing PLC-style input scanning, logic execution, timing control, sequential operations, fault handling, and status monitoring. Therefore, the proposed Arduino-based Virtual PLC aims to provide an economical, flexible, and educational alternative to conventional PLC systems while maintaining essential automation functionalities.

Gap Identification

1. High Cost of Conventional PLCs

Traditional PLC systems are expensive, making them difficult to adopt in educational institutions, research laboratories, and small-scale automation projects where budget constraints are significant.

2. Dependence on Proprietary Hardware and Software

Most industrial PLCs require vendor-specific hardware and licensed programming software, which increases implementation costs and limits flexibility.

2. Limited Accessibility for Learning and Training

Commercial PLC systems are not easily accessible to students and beginners due to their high cost and specialized programming environments.

4. Lack of Affordable Automation Solutions

Many small industries and prototype developers require low-cost automation systems, but existing PLC-based solutions are often economically unfeasible for such applications.

5. Underutilization of Arduino in PLC Applications

Although Arduino provides sufficient processing capability, digital I/O support, and easy programming, its potential as a Virtual PLC platform has not been fully explored in automation systems.

6. Limited PLC-Like Functionality in Existing Arduino Projects

Most Arduino automation projects focus only on basic relay control and do not implement complete PLC features such as input scanning, logic execution, timing operations, fault handling, and sequential control.

7. Complex Expansion in Conventional Automation Systems

Traditional PLC systems often require additional hardware modules for expansion, increasing system complexity and cost.

PROBLEM STATEMENT

Industrial automation systems widely depend on Programmable Logic Controllers (PLCs) for controlling machines, monitoring processes, and ensuring reliable operation. Although PLCs provide excellent performance and durability, their high cost, proprietary software, and specialized hardware make them unsuitable for educational institutions, research laboratories, and small-scale automation applications. Many students and small industries face difficulties in accessing PLC technology due to budget limitations and the complexity of industrial PLC systems.

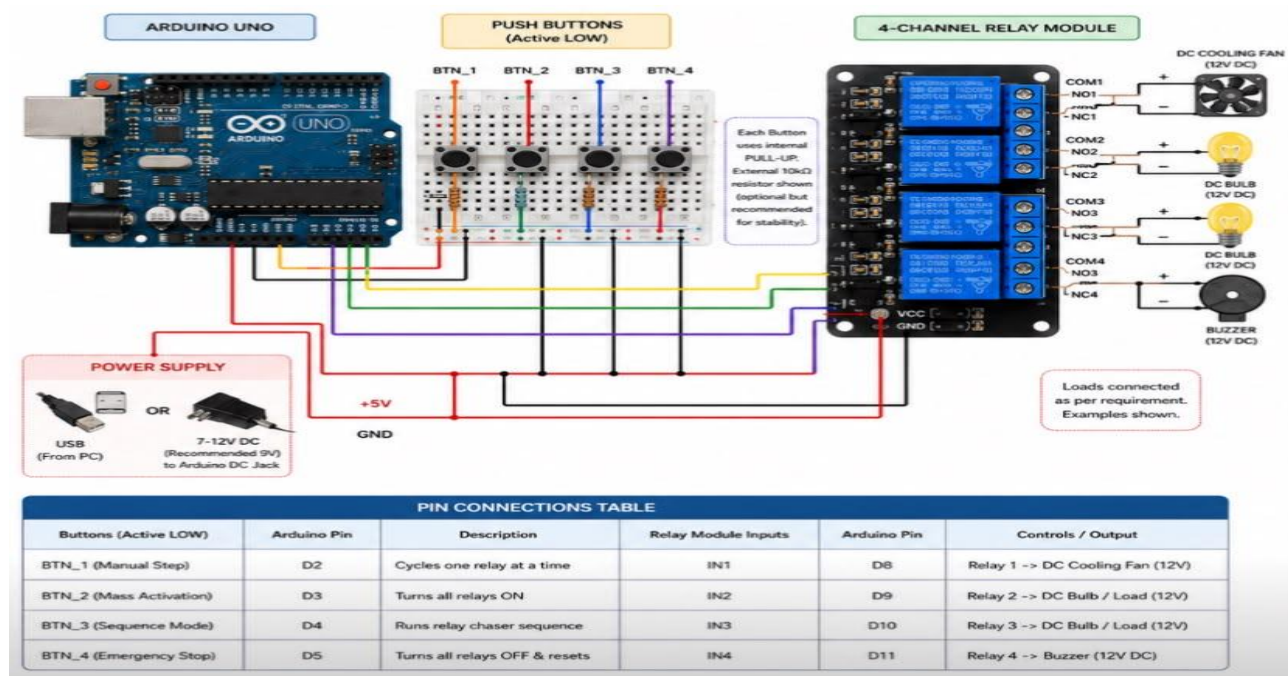
With the advancement of embedded systems, Arduino microcontrollers offer a low-cost, flexible, and easy-to-program platform for automation applications. However, most Arduino-based projects are limited to basic switching and control functions and do not fully replicate the operational characteristics of a PLC. There is a need for a system that can perform essential PLC functions such as input monitoring, logic execution, timing control, sequential operation, and output switching while maintaining simplicity and affordability.

The problem addressed in this project is the design and development of an Arduino-based Virtual PLC capable of emulating the fundamental functions of a conventional PLC. The system should support Manual and Automatic modes of operation, provide real-time status indication, ensure safe control of electrical loads through relays, and offer a user-friendly interface for operation. The proposed solution aims to bridge the gap between traditional PLCs and low-cost embedded controllers by providing an economical, flexible, and educational automation platform suitable for learning, research, and small-scale industrial applications.

OBJECTIVES

- To design and develop a Virtual PLC using the Arduino microcontroller as a low-cost alternative to conventional PLC systems.
- To implement PLC-like control operations such as input scanning, logic execution, timing functions, and output control.
- To provide Manual and Automatic modes of operation for flexible control of electrical loads.
- To control multiple loads through a relay module and demonstrate basic industrial automation concepts.
- To develop a user-friendly interface using push buttons for Start, Stop, Mode Selection, and Reset operations.

System Architecture and Hardware Design



The system architecture of the proposed **Arduino-based Virtual PLC** is designed to emulate the operation of a conventional Programmable Logic Controller (PLC) while maintaining simplicity, low cost, and ease of implementation. The complete system consists of four major sections: the **Power Supply Unit**, **Input Section**, **Control Unit**, and **Output Section**. These sections work together to perform industrial automation functions such as input monitoring, logic execution, timing control, and output switching.

A regulated **5V SMPS (Switched Mode Power Supply)** serves as the primary power source for the Arduino controller, relay module, and status indication LEDs. The power supply ensures stable voltage and reliable operation of all hardware components. Proper grounding is maintained throughout the system to improve safety and reduce electrical noise.

The **Input Section** consists of push-button switches used for Start, Stop, Mode Selection, and Reset operations. These push buttons are connected to the digital input pins of the Arduino through pull-down or pull-up resistor arrangements. The Start button initiates system operation, the Stop button halts all activities, the Mode button allows switching between Manual and Automatic modes, and the Reset button clears any fault conditions. This section functions similarly to the digital input module of a conventional PLC.

The **Control Unit** is implemented using an Arduino Uno microcontroller, which acts as the central processing unit of the Virtual PLC. The Arduino continuously scans the input signals, executes the programmed logic, performs timing functions, monitors system status, and updates output devices. The controller follows a PLC-like scan cycle consisting of input reading, logic processing, fault detection, and output updating. Due to its simple programming environment and sufficient processing capability, Arduino provides a flexible platform for automation applications.

The system also incorporates **status indication LEDs** to provide real-time feedback regarding system conditions. Separate LEDs are used to indicate power availability, operating mode (Manual or Automatic), system running status, and fault conditions. These indicators help users monitor system operation and simplify troubleshooting.

The **Output Section** consists of a four-channel relay module interfaced with the Arduino. Each relay is controlled by a dedicated digital output pin and is used to switch external electrical loads such as lamps, motors, pumps, or other industrial devices. The relay module provides electrical isolation between the low-voltage Arduino circuit and high-voltage load circuits, ensuring safe operation.

In **Manual Mode**, all relay outputs can be activated or deactivated directly according to user commands. In **Automatic Mode**, the Arduino executes a sequential control algorithm where relay outputs are energized one after another with predefined time delays, simulating industrial automation processes.

The hardware design integrates the Arduino controller, push-button inputs, relay module, status LEDs, and regulated power supply into a compact and efficient automation platform. The modular architecture simplifies maintenance, improves reliability, and allows future expansion through the addition of sensors, displays, communication modules, Human Machine Interfaces (HMI), and IoT-based monitoring systems. Thus, the proposed Arduino-based Virtual PLC provides a practical, economical, and flexible alternative to conventional PLC systems for educational, research, and small-scale industrial automation applications.

Working Principle of the System

The working principle of the **Arduino-based Virtual PLC** is based on the fundamental operation of a conventional PLC, which continuously performs input scanning, logic execution, and output updating. When the system is powered ON, a regulated 5V DC supply energizes the Arduino controller, relay module, and status indication LEDs. The power LED glows to indicate that the system is active and ready for operation. The Arduino initializes all input and output pins and keeps all relay outputs in the OFF state to ensure safe startup conditions.

The system continuously monitors the input section, which consists of push buttons used for **Start, Stop, Mode Selection, and Reset** operations. When the Start button is pressed, the Arduino places the system in the run state and begins executing the programmed control logic. The Stop button immediately stops system operation and turns OFF all relay outputs, ensuring safe shutdown of connected loads. The Mode button allows the user to switch between Manual and Automatic modes of operation, while the Reset button clears any fault conditions and restores the system to its initial state.

In **Manual Mode**, the Arduino activates or deactivates all relay outputs simultaneously according to user commands. This mode is useful for applications where all connected loads are required to operate together. In **Automatic Mode**, the Arduino executes a time-based sequential control algorithm. The relay outputs are energized one after another with predefined time delays, simulating industrial process sequences commonly used in conveyor systems, motor control circuits, and automated production lines.

During operation, the Arduino continuously updates the status LEDs to indicate the current condition of the system. Separate LEDs display power status, selected operating mode, system running condition, and fault status. If a fault condition is detected or simulated, the Arduino immediately deactivates all relay outputs and turns ON the fault indicator LED to prevent damage to connected equipment. The fault can only be cleared by pressing the Reset button.

The relay module provides electrical isolation between the low-voltage Arduino control circuit and high-voltage load circuit, ensuring safe and reliable operation. Throughout the process, the Arduino repeatedly performs input monitoring, logic processing, timing control, fault checking, and output updating in a continuous scan cycle. Thus, the Arduino successfully performs PLC-like functions such as input processing, logical decision-making, sequential control, timing operations, and output switching, making it a practical, low-cost, and flexible solution for educational, research, and small-scale industrial automation applications.

Applications of Virtual PLC Using Arduino

1. Industrial Automation

Used for controlling motors, pumps, conveyors, solenoid valves, and other industrial equipment through relay outputs.

2. Process Control Systems

Suitable for sequential operations in manufacturing processes, assembly lines, and packaging industries where operations must occur in a predefined order.

3. Motor Starter and Pump Control

Can automate the starting and stopping of motors and pump sets with safety interlocks and timing control.

4. Smart Lighting Systems

Used for automatic and manual control of lighting in factories, offices, educational institutions, and commercial buildings.

5. Home Automation

Controls household appliances, lighting, fans, and water pumps through programmable logic and wireless connectivity.

6. Agricultural Automation

Useful for irrigation systems, greenhouse automation, water-level monitoring, and pump sequencing in farms.

7. Educational Laboratories

Provides a low-cost platform for students to learn PLC programming, industrial automation, relay interfacing, and embedded systems.

8. IoT-Based Monitoring and Control

The Arduino built-in Wi-Fi enables remote monitoring, data collection, and control through web dashboards and cloud platforms.

9. Building Management Systems

Can be integrated for controlling HVAC systems, lighting, security systems, and energy management in smart buildings.

ADVANTAGES

1. Low Cost

The Arduino-based Virtual PLC is much cheaper than conventional industrial PLCs because it uses an inexpensive Arduino microcontroller and readily available electronic components. This reduces the overall system cost, making it suitable for educational projects, research work, and small-scale automation applications where budget is a major concern.

2. Compact Size

The Arduino controller and relay module occupy very little space compared to traditional PLC systems. Its compact design allows easy installation in small control panels and portable automation setups. The reduced size also simplifies wiring and improves system portability.

3. Easy Programming

The system can be programmed using user-friendly platforms such as Arduino IDE, making it easy to develop, modify, and troubleshoot control logic without requiring specialized PLC programming software.

4. Wireless Connectivity

The built-in Wi-Fi and Bluetooth features of the Arduino enable remote monitoring and control of the system, eliminating the need for additional communication modules and supporting IoT-based automation applications.

5. Flexible Operation

The availability of both Manual and Automatic modes allows the system to adapt to different operational requirements, providing greater flexibility in controlling connected loads and processes.

6. Quick Fault Identification

Status indicator LEDs help operators quickly identify power, run, mode, and fault conditions, reducing troubleshooting time.

7. Scalable Architecture

The system architecture can be expanded to include sensors, HMI interfaces, data logging, and industrial communication protocols for advanced automation applications.

Limitations

1. Not Suitable for Heavy Industrial Applications

The Arduino lacks the ruggedness and certification required for operation in harsh industrial environments with high temperatures, dust, vibration, and electrical noise.

2. Limited Input/Output Capacity

The number of GPIO pins available on the Arduino restricts the number of sensors and actuators that can be connected directly.

3. Lack of Industrial-Grade Isolation

Unlike commercial PLCs, the input circuits do not provide complete electrical isolation, making the system more susceptible to noise and voltage fluctuations.

4. Limited Reliability for Critical Systems

The system does not include advanced safety features such as redundant processors, fail-safe mechanisms, or safety-certified components.

5. Dependence on Software Stability

Any programming error, memory overflow, or software crash can affect system performance and control operations.

6. Relay Wear and Tear

Mechanical relays have a limited switching life and may require replacement after prolonged operation, especially in high-frequency switching applications.

7. Power Supply Sensitivity

Voltage fluctuations or unstable power supplies can cause unexpected resets or malfunction of the Arduino controller.

8. Limited Real-Time Performance

Although Arduino is fast, it cannot match the deterministic real-time performance of dedicated industrial PLC processors in complex control systems.

9. Cybersecurity Risks

When Wi-Fi connectivity is enabled for remote monitoring, the system becomes vulnerable to unauthorized access and network-based attacks if proper security measures are not implemented.

Results

The Arduino-based Virtual PLC was successfully designed, implemented, and tested for controlling four relay outputs in both Manual and Automatic modes. The system effectively performed PLC-like functions such as input scanning, logic execution, timing control, and output switching. In Manual mode, all connected loads operated simultaneously based on user commands, while in Automatic mode, the loads were activated sequentially with predefined time delays. The relay module reliably controlled external electrical loads, and the status LEDs accurately indicated power, operating mode, run condition, and fault status. The system demonstrated stable and responsive performance during testing, validating the capability of the Arduino to function as a low-cost Virtual PLC. Overall, the project achieved its objectives by providing a compact, flexible, and economical automation solution suitable for educational, research, and small-scale industrial applications.

- The Arduino-based Virtual PLC was successfully designed and implemented for controlling four relay outputs using PLC-style logic execution.
- The system effectively operated in Manual Mode, where all connected loads could be switched ON and OFF simultaneously through user commands.
- The Automatic Mode functioned correctly by activating relay outputs sequentially with predefined time delays, simulating industrial process sequencing.
- The Arduino continuously scanned input signals and executed control logic in real time, demonstrating behavior similar to a conventional PLC scan cycle.
- Push-button controls for Start, Stop, Mode Selection, and Reset responded accurately and provided reliable system operation.
- Status indicator LEDs successfully displayed system conditions such as power availability, operating mode, run status, and fault status, improving monitoring capability.
- The relay module provided safe switching of external electrical loads while maintaining isolation between low-voltage control circuits and high-voltage loads.
- Fault handling functionality was implemented successfully, where all relay outputs were turned OFF during fault conditions and restored only after a reset operation.
- The developed system demonstrated a low-cost, compact, and flexible alternative to conventional PLCs, making it suitable for educational, laboratory, and small-scale automation applications.

Conclusion

The Virtual PLC Using Arduino project successfully demonstrates the implementation of a low-cost and efficient automation system capable of performing the basic functions of a conventional PLC. By integrating the Arduino microcontroller, relay modules, push-button inputs, and status indication LEDs, the system effectively executes control logic, sequential operations, and load switching in both Manual and Automatic modes. The project proves that modern embedded controllers can be used as flexible alternatives to traditional PLCs for small-scale automation applications. The system offers advantages such as compact size, easy programmability, wireless connectivity, and expandability while maintaining reliable performance. Although it is not intended to replace industrial-grade PLCs in critical applications, it serves as an excellent platform for education, research, prototyping, and basic industrial automation. Overall, the project successfully achieves its objectives and provides a strong foundation for future developments in IoT-based and smart automation systems.

Future Scope

1. Integration of IoT Technology

The system can be connected to cloud platforms for remote monitoring, control, and data analysis through the internet.

2. Development of Mobile Applications

A dedicated Android or iOS application can be developed for real-time monitoring and control of the PLC system.

3. Human Machine Interface (HMI) Integration

An HMI display can be added to provide a graphical interface for easier operation and visualization of system status.

4. Implementation of Industrial Communication Protocols

Protocols such as Modbus, MQTT, Ethernet/IP, and OPC-UA can be incorporated for industrial networking and communication.

5. Addition of Advanced Sensors

Temperature, pressure, proximity, current, and level sensors can be integrated for intelligent monitoring and control.

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